

Multi-Turn Speech Dialogue

A curated paper list for Multi-Turn Speech Dialogue.

Survey

1. "Turn-taking in Conversational Systems and Human-Robot Interaction: A Review". *Gabriel Skantze* Computer Speech & Language 2021. [Paper](#)
2. "A Review of Dialogue Systems: From Trained Monkeys to Stochastic Parrots". *Atharv Singh Patlan et al.* arXiv 2023. [Paper](#)
3. "WavChat: A Survey of Spoken Dialogue Models". *Shengpeng Ji et al.* arXiv 2024. [Paper](#)
4. "From Turn-Taking to Synchronous Dialogue: A Survey of Full-Duplex Spoken Language Models". *Yuxuan Chen and Haoyuan Yu* arXiv 2025. [Paper](#)
5. "A Survey of Recent Advances on Turn-taking Modeling in Spoken Dialogue Systems". *Galo Castillo-López et al.* IWSDS 2025. [Paper](#)
6. "Turn-Taking Modelling in Conversational Systems: A Review of Recent Advances". *Rutherford Agbeshi Patamia et al.* Technologies 2025. [Paper](#)
7. "A Survey of Full-Duplex Spoken Dialogue Systems: Architectural Hierarchy, Interaction Ontology, and Decision State Machine". *Jingyu Lu et al.* arXiv 2026. [Paper](#)

General Surveys

1. "A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version". *Iulian V. Serban et al.* Dialogue & Discourse 2018. [Paper](#)
2. "Neural Approaches to Conversational AI". *Jianfeng Gao et al.* Foundations and Trends in Information Retrieval 2019. [Paper](#)
3. "Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey". *Vevake Balaraman et al.* SIGDIAL 2021. [Paper](#)
4. "Who Says What to Whom: A Survey of Multi-Party Conversations". *Jia-Chen Gu et al.* IJCAI 2022. [Paper](#)
5. "Transformers in Speech Processing: A Survey". *Siddique Latif et al.* arXiv 2023. [Paper](#)
6. "Preference Tuning with Human Feedback on Language, Speech, and Vision Tasks: A Survey". *Genta Indra Winata et al.* arXiv 2024. [Paper](#)
7. "A Survey on Speech Large Language Models". *Jing Peng et al.* arXiv 2024. [Paper](#)
8. "On The Landscape of Spoken Language Models: A Comprehensive Survey". *Siddhant Arora et al.* arXiv 2025. [Paper](#)
9. "Recent Advances in Speech Language Models: A Survey". *Wenqian Cui et al.* ACL 2025. [Paper](#)
10. "Aligning Multimodal LLM with Human Preference: A Survey". *Tao Yu et al.* arXiv 2025. [Paper](#)

Datasets

1. "Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags". *Koichiro Yoshino et al.*. LREC 2018. [Paper](#)
2. "Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset". *Bill Byrne et al.*. EMNLP-IJCNLP 2019. [Paper](#)
3. "Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment". *Benjamin Benk et al.*. LREC 2020. [Paper](#)
4. "HarperValleyBank: A Domain-Specific Spoken Dialog Corpus". *Mike Wu et al.*. arXiv 2020. [Paper](#)
5. "The MSP-Conversation Corpus". *Luz Martinez-Lucas et al.*. Interspeech 2020. [Paper](#)
6. "ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation". *Holy Lovenia et al.*. arXiv 2021. [Paper](#)
7. "DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech". *Keon Lee et al.* arXiv 2022. [Paper](#)
8. "DOROTHIE: Spoken Dialogue for Handling Unexpected Situations in Interactive Autonomous Driving Agents". *Ziqiao Ma et al.* EMNLP Findings 2022. [Paper](#)
9. "Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers". *Michimasa Inaba et al.*. LREC 2022. [Paper](#)
10. "Design and Evaluation of the Corpus of Everyday Japanese Conversation". *Hanae Koiso et al.*. LREC 2022. [Paper](#)
11. "SpokenWOZ: A Large-Scale Speech-Text Benchmark for Spoken Task-Oriented Dialogue Agents". *Shuzheng Si et al.* arXiv 2023. [Paper](#)
12. "CALLS: Japanese Empathetic Dialogue Speech Corpus of Complaint Handling and Attentive Listening in Customer Center". *Yuki Saito et al.* Interspeech 2023. [Paper](#)
13. "Advancing Large Language Models to Capture Varied Speaking Styles and Respond Properly in Spoken Conversations (StyleTalk)". *Guan-Ting Lin et al.* ACL 2024. [Paper](#)
14. "Generative Expressive Conversational Speech Synthesis (NCSSD)". *Rui Liu et al.* ACM MM 2024. [Paper](#)
15. "J-CHAT: Japanese Large-scale Spoken Dialogue Corpus for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2024. [Paper](#)
16. "DeepDialogue: A Multi-Turn Emotionally-Rich Spoken Dialogue Dataset". *Alkis Koudounas et al.* arXiv 2025. [Paper](#)
17. "UltraVoice: Scaling Fine-Grained Style-Controlled Speech Conversations for Spoken Dialogue Models". *Wenming Tu et al.* arXiv 2025. [Paper](#)
18. "Toward Conversational Hungarian Speech Recognition: Introducing the BEA-Large and BEA-Dialogue Datasets". *Mate Gedeon et al.* arXiv 2025. [Paper](#)
19. "InteractSpeech: A Speech Dialogue Interaction Corpus for Spoken Dialogue Model". *Yifu Chen et al.* EMNLP Findings 2025. [Paper](#)
20. "Behavior-SD: Behaviorally Aware Spoken Dialogue Generation with Large Language Models". *Sehun Lee et al.* NAACL 2025. [Paper](#)

21. "RealTalk-CN: A Realistic Chinese Speech-Text Dialogue Benchmark With Cross-Modal Interaction Analysis". *Enzhi Wang et al.* arXiv 2025. [Paper](#)
22. "DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue". *Zheyuan Zhang et al.* arXiv 2025. [Paper](#)
23. "Open-Source Full-Duplex Conversational Datasets for Natural and Interactive Speech Synthesis". *Rui Liu et al.* arXiv 2025. [Paper](#)
24. "Exploring Emotional Nuances in Spoken Dialogue: Dataset Construction and Prediction of Emotional Dialogue Breakdown". *Hyuga Nakaguro et al.* IWSDS 2026. [Paper](#)
25. "DuplexChat: Constructing Speaker-Separated Full-Duplex Dialogue Speech at Scale for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2026. [Paper](#)
26. "Dial HEALTHDIAL for Advice: A Multilingual and Multi-Parallel Spoken Dialogue Dataset for Knowledge-Grounded Information Seeking". *Songbo Hu et al.* ACL Findings 2026. [Paper](#)

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. "Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs". *Matthew Roddy et al.* Interspeech 2018. [Paper](#)
2. "Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network". *Nigel G. Ward et al.* SLT 2018. [Paper](#)
3. "How Much Does Prosody Help Turn-taking? Investigations using Voice Activity Projection Models". *Erik Ekstedt et al.* SIGDIAL 2022. [Paper](#)
4. "Talking Turns: Benchmarking Audio Foundation Models on Turn-Taking Dynamics". *Siddhant Arora et al.* ICLR 2025. [Paper](#)
5. "Investigating the Impact of Incremental Processing and Voice Activity Projection on Spoken Dialogue Systems". *Yuya Chiba et al.* COLING 2025. [Paper](#)
6. "Synchronization and Turn-Taking in Full-Duplex Speech Dialogue Models". *Pablo Riera et al.* arXiv 2026. [Paper](#)
7. "Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions". *Dongwook Lee et al.* ACL 2026. [Paper](#)
8. "Rethinking Binary Evaluation of Turn-Taking under Inherent Ambiguity". *Yunosuke Kubo et al.* SIGDIAL 2026. [Paper](#)

Evaluation Focus: Full-duplex Interaction

1. "MMedFD: A Real-world Healthcare Benchmark for Multi-turn Full-Duplex Automatic Speech Recognition". *Hongzhao Chen et al.* arXiv 2025. [Paper](#)
2. "Full-Duplex-Bench: A Benchmark to Evaluate Full-duplex Spoken Dialogue Models on Turn-taking Capabilities". *Guan-Ting Lin et al.* arXiv 2025. [Paper](#)
3. "FD-Bench: A Full-Duplex Benchmarking Pipeline Designed for Full Duplex Spoken Dialogue Systems". *Yizhou Peng et al.* Interspeech 2025. [Paper](#)

4. "The ICASSP 2026 HumDial Challenge: Benchmarking Human-like Spoken Dialogue Systems in the LLM Era". *Zhixian Zhao et al.* ICASSP / arXiv 2026. [Paper](#)
5. "T-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains". *Soham Ray et al.* ICML 2026. [Paper](#)
6. "Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency". *Guan-Ting Lin et al.* arXiv 2026. [Paper](#)
7. "Full-Duplex Interaction in Spoken Dialogue Systems: A Comprehensive Study from the ICASSP 2026 HumDial Challenge". *Chengyou Wang et al.* arXiv 2026. [Paper](#)
8. "Full-Duplex-Bench-v2: A Multi-Turn Evaluation Framework for Duplex Dialogue Systems with an Automated Examiner". *Guan-Ting Lin et al.* ACL 2026. [Paper](#)
9. "MTR-DuplexBench: Towards a Comprehensive Evaluation of Multi-Round Conversations for Full-Duplex Speech Language Models". *Zhang He et al.* ACL Findings 2026. [Paper](#)
10. "M3-DuplexBench: A Multi-Turn, Multilingual, Multidomain Benchmark for Full-Duplex Spoken Dialogue Models". *Ryo Fukuda et al.* arXiv 2026. [Paper](#)

Evaluation Focus: Multi-turn Dialogue Capabilities

1. "SD-Eval: A Benchmark Dataset for Spoken Dialogue Understanding Beyond Words". *Junyi Ao et al.* NeurIPS 2024 (Datasets & Benchmarks). [Paper](#)
2. "Contextual Interactive Evaluation of TTS Models in Dialogue Systems". *Siyang Wang et al.* Interspeech 2024. [Paper](#)
3. "Benchmarking Open-ended Audio Dialogue Understanding for Large Audio-Language Models". *Kuofeng Gao et al.* arXiv 2024. [Paper](#)
4. "URO-Bench: Towards Comprehensive Evaluation for End-to-End Spoken Dialogue Models". *Ruiqi Yan et al.* EMNLP Findings 2025. [Paper](#)
5. "WavReward: Spoken Dialogue Models With Generalist Reward Evaluators". *Shengpeng Ji et al.* arXiv 2025. [Paper](#)
6. "MAD: A Benchmark for Multi-Turn Audio Dialogue Fact-Checking". *Chaewan Chun et al.* arXiv 2025. [Paper](#)
7. "VoxRole: A Comprehensive Benchmark for Evaluating Speech-Based Role-Playing Agents". *Weihao Wu et al.* arXiv 2025. [Paper](#)
8. "Evaluating Bias in Spoken Dialogue LLMs for Real-World Decisions and Recommendations (FairDialogue)". *Yihao Wu et al.* arXiv 2025. [Paper](#)
9. "MULTI-Bench: A Multi-turn Interactive Benchmark for Assessing Emotional Intelligence Ability of Spoken Dialogue Models". *Yayue Deng et al.* arXiv 2025. [Paper](#)
10. "Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction". *Advait Gosai et al.* arXiv 2025. [Paper](#)
11. "VoxDialogue: Can Spoken Dialogue Systems Understand Information Beyond Words?". *Xize Cheng et al.* ICLR 2025. [Paper](#)
12. "EchoMind: An Interrelated Multi-level Benchmark for Evaluating Empathetic Speech Language Models". *Li Zhou et al.* arXiv 2025. [Paper](#)
13. "SOVA-Bench: Benchmarking the Speech Conversation Ability for LLM-based Voice Assistant". *Yixuan Hou et al.* Interspeech 2025. [Paper](#)

14. "MTalk-Bench: Evaluating Speech-to-Speech Models in Multi-Turn Dialogues via Arena-style and Rubrics Protocols". *Yuhao Du et al.* arXiv 2025. [Paper](#)
15. "VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?". *Dhruv Jain et al.* arXiv 2025. [Paper](#)
16. "C3: A Bilingual Benchmark for Spoken Dialogue Models Exploring Challenges in Complex Conversations". *Chengqian Ma et al.* EMNLP 2025. [Paper](#)
17. "HumDial-EIBench: A Human-Recorded Multi-Turn Emotional Intelligence Benchmark for Audio Language Models". *Shuiyuan Wang et al.* arXiv 2026. [Paper](#)
18. "VoiceBench: Benchmarking LLM-Based Voice Assistants". *Yiming Chen et al.* TACL 2026. [Paper](#)
19. "The Context Trap: Why End-to-End Audio Language Models Fail Multi-turn Dialogues". *Zhi Rui Tam et al.* IWSDS 2026. [Paper](#)
20. "SDiaReward: Modeling and Benchmarking Spoken Dialogue Rewards with Modality and Colloquialness". *Jingyu Lu et al.* ACL 2026. [Paper](#)
21. "SpeakerSleuth: Can Large Audio-Language Models Judge Speaker Consistency across Multi-turn Dialogues?". *Jonggeun Lee et al.* ACL 2026. [Paper](#)
22. "Style Amnesia: Investigating Speaking Style Degradation and Mitigation in Multi-Turn Spoken Language Models". *Yu-Xiang Lin et al.* ACL Findings 2026. [Paper](#)

Models

Interaction Mode: Turn-taking

1. "Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task". *Zakaria Aldeneh et al.* ICASSP 2018. [Paper](#)
2. "Neural Dialogue Context Online End-of-Turn Detection". *Ryo Masumura et al.* SIGDIAL 2018. [Paper](#)
3. "Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers". *Kohei Hara et al.* Interspeech 2018. [Paper](#)
4. "Turn-Taking Prediction Based on Detection of Transition Relevance Place". *Kohei Hara et al.* Interspeech 2019. [Paper](#)
5. "TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog". *Erik Ekstedt and Gabriel Skantze.* EMNLP Findings 2020. [Paper](#)
6. "Projection of Turn Completion in Incremental Spoken Dialogue Systems". *Erik Ekstedt and Gabriel Skantze.* SIGDIAL 2021. [Paper](#)
7. "Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation". *Shinya Fujie et al.* Interspeech 2021. [Paper](#)
8. "Using Transition Duration to Improve Turn-taking in Conversational Agents". *Charles Threlkeld et al.* SIGDIAL 2022. [Paper](#)
9. "When can I Speak? Predicting Initiation Points for Spoken Dialogue Agents". *Siyan Li et al.* SIGDIAL 2022. [Paper](#)

10. "Voice Activity Projection: Self-supervised Learning of Turn-taking Events". *Erik Ekstedt and Gabriel Skantze* Interspeech 2022. [Paper](#)
11. "Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue". *Jiudong Yang et al.* ICASSP 2022. [Paper](#)
12. "Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation". *Jin Sakuma et al.* Interspeech 2022. [Paper](#)
13. "Turn-Taking Prediction for Natural Conversational Speech". *Shuo-Yiin Chang et al.* Interspeech 2022. [Paper](#)
14. "SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities". *Dong Zhang et al.* EMNLP Findings 2023. [Paper](#)
15. "Generative Spoken Dialogue Language Modeling". *Tu Anh Nguyen et al.* TACL 2023. [Paper](#)
16. "Let's Go Real Talk: Spoken Dialogue Model for Face-to-Face Conversation". *Se Park et al.* ACL 2024. [Paper](#)
17. "Qwen2-Audio Technical Report". *Yunfei Chu et al.* arXiv 2024. [Paper](#)
18. "Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation". *Yinghao Aaron Li et al.* arXiv 2024. [Paper](#)
19. "EMOVA: Empowering Language Models to See, Hear and Speak with Vivid Emotions". *Kai Chen et al.* arXiv 2024. [Paper](#)
20. "Internalizing ASR with Implicit Chain of Thought for Efficient Speech-to-Speech Conversational LLM". *Robin Shinghe Yuen et al.* NeurIPS 2024. [Paper](#)
21. "Building a Taiwanese Mandarin Spoken Language Model: A First Attempt". *Chih-Kai Yang et al.* arXiv 2024. [Paper](#)
22. "LLaMA-Omni: Seamless Speech Interaction with Large Language Models". *Qingkai Fang et al.* arXiv 2024. [Paper](#)
23. "Mini-Omni: Language Models Can Hear, Talk While Thinking in Streaming". *Zhifei Xie et al.* arXiv 2024. [Paper](#)
24. "GLM-4-Voice: Towards Intelligent and Human-Like End-to-End Spoken Chatbot". *Aohan Zeng et al.* arXiv 2024. [Paper](#)
25. "Triadic Multi-party Voice Activity Projection for Turn-taking in Spoken Dialogue Systems". *Mikey Elmers et al.* Interspeech 2025. [Paper](#)
26. "OpenOmni: Advancing Open-Source Omnimodal Large Language Models with Progressive Multimodal Alignment and Real-time Emotional Speech Synthesis". *Run Luo et al.* arXiv 2025. [Paper](#)
27. "GOAT-SLM: A Spoken Language Model with Paralinguistic and Speaker Characteristic Awareness". *Hongjie Chen et al.* arXiv 2025. [Paper](#)
28. "OpenS2S: Advancing Fully Open-Source End-to-End Empathetic Large Speech Language Model". *Chen Wang et al.* arXiv 2025. [Paper](#)
29. "Step-Audio 2 Technical Report". *StepFun Team et al.* arXiv 2025. [Paper](#)
30. "InteractiveOmni: A Unified Omni-modal Model for Audio-Visual Multi-turn Dialogue". *Wenwen Tong et al.* arXiv 2025. [Paper](#)

31. "Step-Audio: Unified Understanding and Generation in Intelligent Speech Interaction". *Ailin Huang et al.* arXiv 2025. [Paper](#)
32. "Baichuan-Audio: A Unified Framework for End-to-End Speech Interaction". *Tianpeng Li et al.* arXiv 2025. [Paper](#)
33. "Kimi-Audio Technical Report". *KimiTeam et al.* arXiv 2025. [Paper](#)
34. "Qwen3-Omni Technical Report". *Jin Xu et al.* arXiv 2025. [Paper](#)
35. "PACHAT: Persona-Aware Speech Assistant for Multi-party Dialogue". *Dongjie Fu et al.* EMNLP 2025. [Paper](#)
36. "VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large Speech-Language Model". *Zuwei Long et al.* arXiv 2025. [Paper](#)
37. "AV-Dialog: Spoken Dialogue Models with Audio-Visual Input". *Tuochao Chen et al.* arXiv 2025. [Paper](#)
38. "Qwen2.5-Omni Technical Report". *Qwen Team et al.* arXiv 2025. [Paper](#)
39. "LLaMA-Omni 2: LLM-based Real-time Spoken Chatbot with Autoregressive Streaming Speech Synthesis". *Qingkai Fang et al.* ACL 2025. [Paper](#)
40. "VoxMind: An End-to-End Agentic Spoken Dialogue System". *Tianle Liang et al.* ACL 2026. [Paper](#)
41. "ZipVoice-Dialog: Non-Autoregressive Spoken Dialogue Generation with Flow Matching". *Han Zhu et al.* ACL Findings 2026. [Paper](#)

Interaction Mode: Half-duplex / Controlled Barge-in

1. "An Incremental Turn-Taking Model for Task-Oriented Dialog Systems". *Andrei C. Coman et al.* Interspeech 2019. [Paper](#)
2. "Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions". *Daniel Ortega et al.* ICASSP 2020. [Paper](#)
3. "BPM_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning". *Jin Yea Jang et al.* EMNLP 2021. [Paper](#)
4. "Device Directedness with Contextual Cues for Spoken Dialog Systems". *Dhanush Bekal et al.* arXiv 2022. [Paper](#)
5. "Contextual Acoustic Barge-In Classification for Spoken Dialog Systems". *Dhanush Bekal et al.* Interspeech 2022. [Paper](#)
6. "Freeze-Omni: A Smart and Low Latency Speech-to-Speech Dialogue Model with Frozen LLM". *Xiong Wang et al.* arXiv 2024. [Paper](#)
7. "Mini-Omni2: Towards Open-source GPT-4o with Vision, Speech and Duplex Capabilities". *Zhifei Xie et al.* arXiv 2024. [Paper](#)
8. "VITA: Towards Open-Source Interactive Omni Multimodal LLM". *Chaoyou Fu et al.* arXiv 2024. [Paper](#)
9. "SHANKS: Simultaneous Hearing and Thinking for Spoken Language Models". *Cheng-Han Chiang et al.* arXiv 2025. [Paper](#)

Interaction Mode: Full-duplex

1. "Duplex Conversation in Outbound Agent System". *Chunxiang Jin et al.* Interspeech 2021. [Paper](#)

2. "Duplex Conversation: Towards Human-like Interaction in Spoken Dialogue Systems". *Ting-En Lin et al.* arXiv 2022. [Paper](#)
3. "Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents". *Bandhav Veluri et al.* EMNLP 2024. [Paper](#)
4. "Moshi: A Speech-Text Foundation Model for Real-Time Dialogue". *Alexandre Defossez et al.* arXiv 2024. [Paper](#)
5. "SALMONN-omni: A Codec-free LLM for Full-duplex Speech Understanding and Generation". *Wenyi Yu et al.* arXiv 2024. [Paper](#)
6. "A Full-duplex Speech Dialogue Scheme Based On Large Language Models". *Peng Wang et al.* NeurIPS 2024. [Paper](#)
7. "MinMo: A Multimodal Large Language Model for Seamless Voice Interaction". *FunAudioLLM Team et al.* arXiv 2025. [Paper](#)
8. "SALM-Duplex: Efficient and Direct Duplex Modeling for Speech-to-Speech Language Model". *Ke Hu et al.* arXiv 2025. [Paper](#)
9. "OmniFlatten: An End-to-end GPT Model for Seamless Voice Conversation". *Qinglin Zhang et al.* ACL 2025. [Paper](#)
10. "Towards a Japanese Full-duplex Spoken Dialogue System". *Atsumoto Ohashi et al.* Interspeech 2025. [Paper](#)
11. "FireRedChat: A Pluggable, Full-Duplex Voice Interaction System with Cascaded and Semi-Cascaded Implementations". *Junjie Chen et al.* arXiv 2025. [Paper](#)
12. "Incorporating Dialogue State Tracking into Japanese Full-duplex Task-oriented Spoken Dialogue Model". *Yuya Chiba et al.* IJCNLP-AACL 2025. [Paper](#)
13. "Chronological Thinking in Full-Duplex Spoken Dialogue Language Models". *Donghang Wu et al.* arXiv 2025. [Paper](#)
14. "LLM-Enhanced Dialogue Management for Full-Duplex Spoken Dialogue Systems". *Hao Zhang et al.* arXiv 2025. [Paper](#)
15. "Language Model Can Listen While Speaking". *Ziyang Ma et al.* AACL 2025. [Paper](#)
16. "FlashLabs Chroma 1.0: A Real-Time End-to-End Spoken Dialogue Model with Personalized Voice Cloning". *Tanyu Chen et al.* arXiv 2026. [Paper](#)
17. "PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models". *Rajarshi Roy et al.* arXiv 2026. [Paper](#)
18. "MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction". *Junbo Cui et al.* arXiv 2026. [Paper](#)
19. "DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action". *Haoyang Zhang et al.* arXiv 2026. [Paper](#)
20. "BayLing-Duplex: Native Full-Duplex Speech Dialogue with a Single Autoregressive LLM". *Qingkai Fang et al.* arXiv 2026. [Paper](#)
21. "F-Actor: Controllable Conversational Behavior in Full-Duplex Models". *Maike Züfle et al.* ACL Findings 2026. [Paper](#)
22. "Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs". *Zhenyu Liu et al.* ACL 2026. [Paper](#)

23. "JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents". *Yinhao Bai et al.* arXiv 2026. [Paper](#)
24. "SoulX-Duplug: Plug-and-Play Streaming State Prediction Module for Realtime Full-Duplex Speech Conversation". *Ruiqi Yan et al.* arXiv 2026. [Paper](#)

Training Methods

Interaction Mode: Turn-taking

1. "Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations". *Ruijie Yan et al.* arXiv 2022. [Paper](#)
2. "Adapting Text-based Dialogue State Tracker for Spoken Dialogues". *Jaeseok Yoon et al.* DSTC 2023. [Paper](#)
3. "SpeechAlign: Aligning Speech Generation to Human Preferences". *Dong Zhang et al.* NeurIPS 2024. [Paper](#)
4. "Data-Centric Improvements for Enhancing Multi-Modal Understanding in Spoken Conversation Modeling (ASK-QA)". *Maximillian Chen et al.* ACL Findings 2025. [Paper](#)
5. "WavRAG: Audio-Integrated Retrieval Augmented Generation for Spoken Dialogue Models". *Yifu Chen et al.* ACL 2025. [Paper](#)
6. "Enhancing Speech-to-Speech Dialogue Modeling with End-to-End Retrieval-Augmented Generation". *Pengchao Feng et al.* EMNLP Findings 2025. [Paper](#)
7. "Aligning Spoken Dialogue Models from User Interactions". *Anne Wu et al.* arXiv 2025. [Paper](#)
8. "Chain-of-Thought Training for Open E2E Spoken Dialogue Systems". *Siddhant Arora et al.* arXiv 2025. [Paper](#)
9. "VoiceTextBlender: Augmenting Large Language Models with Speech Capabilities via Single-Stage Joint Speech-Text Supervised Fine-Tuning". *Yifan Peng et al.* NAACL 2025. [Paper](#)
10. "Approaching Dialogue State Tracking via Aligning Speech Encoders and LLMs". *Šimon Sedláček et al.* arXiv 2025. [Paper](#)
11. "The Speech-LLM Takes It All: A Truly Fully End-to-End Spoken Dialogue State Tracking Approach". *Nizar El Ghazal et al.* arXiv 2025. [Paper](#)
12. "Optimizing Conversational Quality in Spoken Dialogue Systems with Reinforcement Learning from AI Feedback". *Siddhant Arora et al.* arXiv 2026. [Paper](#)
13. "WavAlign: Enhancing Intelligence and Expressiveness in Spoken Dialogue Models via Adaptive Hybrid Post-Training". *Yifu Chen et al.* ACL Findings 2026. [Paper](#)

Interaction Mode: Half-duplex / Streaming

1. "A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning". *Hatim Khouzaimi et al.* Computer Speech & Language 2018. [Paper](#)
2. "Stream RAG: Instant and Accurate Spoken Dialogue Systems with Streaming Tool Usage". *Siddhant Arora et al.* arXiv 2025. [Paper](#)
3. "Dual-Reasoner: Bridging Interleaved Atomicity and Streaming Latency via Thinking-while-Talking". *Yangzhuo Li et al.* ACL Findings 2026. [Paper](#)

Interaction Mode: Full-duplex

1. "FLM-Audio: Natural Monologues Improves Native Full-Duplex Chatbots via Dual Training". *Yiqun Yao et al.* arXiv 2025. [Paper](#)
2. "Reinforcement Learning Enhanced Full-Duplex Spoken Dialogue Language Models for Conversational Interactions". *Chen Chen et al.* COLM 2025. [Paper](#)
3. "Multi-Faceted Interactivity Alignment in Full-Duplex Speech Models". *Atsumoto Ohashi et al.* arXiv 2026. [Paper](#)
4. "Reproducing Proficiency-Conditioned Dialogue Features with Full-duplex Spoken Dialogue Models". *Takao Obi et al.* IWSDS 2026. [Paper](#)
5. "Effects of Dialogue Corpora Properties on Fine-Tuning a Moshi-Based Spoken Dialogue Model". *Yuto Abe et al.* IWSDS 2026. [Paper](#)
6. "Dual-Axis Generative Reward Model Toward Semantic and Turn-taking Robustness in Interactive Spoken Dialogue Models". *Yifu Chen et al.* ACL 2026. [Paper](#)
7. "Sommelier: Scalable Open Multi-turn Audio Pre-processing for Full-duplex Speech Language Models". *Kyudan Jung et al.* ACL Industry Track 2026. [Paper](#)

Survey Coverage Comparison

Table 1. Coverage comparison of surveys related to Multi-Turn Speech Dialogue.

General Surveys are listed first, followed by core surveys. ✓ = systematic coverage; △ = partial or supporting coverage; - = not a substantive focus.

Survey	Research Scope			Content Coverage				Interaction Taxonomy
	Multi-turn Speech	Turn-taking & Interruption	Full-duplex	Datasets	Evaluation/Bench mark	Models	Training Methods	
Serban et al. (2018)	△	-	-	✓	✓	△	△	✓
Gao et al. (2019)	△	-	-	△	△	✓	✓	✓
Balaraman et al. (2021)	△	-	-	✓	✓	✓	△	✓
Gu et al. (2022)	△	△	-	△	△	✓	△	✓
Latif et al. (2023)	△	-	-	△	△	✓	△	✓
Winata et al. (2024)	-	-	-	✓	✓	△	✓	✓
Peng et al. (2024)	△	-	-	△	✓	✓	✓	✓
Arora et al. (2025)	△	△	△	△	✓	✓	✓	✓
Cui et al. (2025)	△	-	△	△	✓	✓	✓	✓
Yu et al. (2025)	-	-	-	✓	✓	△	✓	✓
Skantze (2021)	✓	✓	△	△	△	✓	-	✓
Patlan et al. (2023)	✓	△	-	△	✓	✓	✓	✓
Ji et al. / WavChat (2024)	✓	✓	✓	✓	✓	✓	✓	✓
Chen and Yu (2025)	✓	✓	✓	△	✓	✓	△	✓
Castillo-López et al. (2025)	✓	✓	-	✓	△	✓	△	✓
Patamia et al. (2025)	✓	✓	△	✓	✓	✓	△	✓
Lu et al. (2026)	✓	✓	✓	✓	✓	✓	△	✓
Current Collection	✓	✓	✓	✓	✓	✓	✓	✓

System and Model Comparison

Scope. Component = predictor/classifier used inside a dialogue pipeline; Model = speech or multimodal foundation model; System = complete dialogue pipeline; Add-on = plug-in module that upgrades another system.

Interaction support. Streaming processes or generates audio incrementally; Barge-in accepts user interruption; Concurrent means listening while speaking; Backchannel/Overlap covers feedback and overlapping speech.

Architecture. Cascaded uses explicit ASR/LLM-or-DM/TTS stages; Semi-cascaded partially integrates the speech path but retains replaceable frontends, decoders, or controllers; End-to-end directly models speech representations without a mandatory transcript as the central interface.

Controller-mediated support. The capability is triggered by an external VAD, endpoint detector, state predictor, scheduler, or dialogue manager rather than decided entirely inside the main generative backbone. Such partial or indirect support is marked △.

Multiple architecture checks. Architecture columns describe implementations reported by the paper, not mutually exclusive labels for the publication. FireRedChat, for example, explicitly provides both cascaded and semi-cascaded implementations.

Table 2. Comparison of all papers listed under Models for Multi-Turn Speech Dialogue.

✓ = explicitly reported support; △ = partial, asymmetric, or controller-mediated support; - = not reported, not applicable, or not a primary capability. Method names link to the papers.

Method	Scope	Interaction Capabilities				Modalities		System Architecture			Speech Interface	Core Design	Release
		Streaming	Barge-in	Concurrent	Backchannel / Overlap	Speech Output	Vision	Cascaded	Semi-casc.	End-to-end			
Interaction Mode: Turn-taking													
EOT + Intent	Component	△	-	-	-	-	-	-	-	-	Acoustic features	End-of-turn detector with auxiliary intention task	2018
Neural Online EOT	Component	✓	-	-	-	-	-	-	-	-	Acoustic + context	Online neural end-of-turn detection	2018
MTL Turn-taking	Component	△	-	-	✓	-	-	-	-	-	Acoustic features	Joint turn-taking, backchannel, and filler prediction	2018
TRP Turn-taking	Component	△	-	-	-	-	-	-	-	-	Speech features	Transition-relevance-place detection	2019
TurnGPT	Component	△	-	-	-	-	-	-	-	-	Text tokens	Transformer language model for turn-shift prediction	2020
Turn Completion Projection	Component	✓	-	-	-	-	-	-	-	-	Incremental text	Incremental projection of turn completion	2021
Timing Generating Networks	Component	△	-	-	-	-	-	-	-	-	Timing features	Neural timing generation for multiparty turns	2021
Transition Duration	Component	△	-	-	-	-	-	-	-	-	Timing features	Transition-duration cue for agent turn-taking	2022
Initiation Point Predictor	Component	△	-	-	-	-	-	-	-	-	Speech + context	Predicts when an agent may initiate speech	2022
VAP	Component	✓	-	-	✓	-	-	-	-	-	Waveform	Self-supervised future voice-activity projection	2022
Gated Multimodal Fusion	Component	△	-	-	-	-	✓	-	-	-	Audio-visual features	Contrastive gated multimodal turn predictor	2022
Response Timing Estimator	Component	△	-	-	-	-	-	-	-	-	Dialogue acts	Response timing conditioned on dialogue acts	2022
Natural TT Predictor	Component	△	-	-	-	-	-	-	-	-	Conversational speech	Turn-taking prediction from natural speech	2022
SpeechGPT	Model	-	-	-	-	✓	✓	-	-	✓	HuBERT units	LLaMA with speech tokenizer and unit vocoder	2023
dGSLM	Model	△	-	△	✓	✓	-	-	-	✓	HuBERT units	Dual-stream textless Transformer dialogue LM	2023
RealTalk	System	△	-	-	-	✓	-	-	-	✓	Speech features	Face-to-face spoken dialogue modeling	2024
Qwen2-Audio	Model	-	-	-	-	-	-	-	✓	-	Whisper features	Audio encoder connected to a text-output LLM	2024
Style-Talker	System	△	-	-	-	-	✓	-	-	✓	Audio LM + TTS	Audio language model with style-conditioned TTS	2024
EMOVA	Model	-	-	-	-	✓	✓	-	-	✓	Semantic + acoustic	Omnimodal LLM with emotion-aware speech generation	2024
IntrinsicVoice	Model	△	-	-	-	✓	-	-	-	✓	HuBERT units	Implicit ASR chain-of-thought with GroupFormer	2024
Taiwanese Mandarin SLM	Model	-	-	-	-	✓	-	-	-	✓	Discrete units	Speech language modeling for Taiwanese Mandarin	2024
LLaMA-Omni	System	△	-	-	-	-	✓	-	-	✓	Whisper + units	Speech adapter, LLM, and streaming speech decoder	2024
Mini-Omni	System	△	-	-	-	✓	-	-	-	✓	SNAC codec	Delayed parallel text-speech token decoding	2024
GLM-4-Voice	System	✓	-	-	-	✓	-	-	-	✓	Speech tokenizer	Unified tokenizer, GLM, and flow speech decoder	2024
Triadic VAP	Component	✓	-	-	✓	-	-	-	-	-	Multi-party waveform	Voice-activity projection for triadic dialogue	2025
OpenOmni	Model	△	-	-	-	✓	✓	-	-	✓	Speech tokens	Progressive multimodal alignment and emotional TTS	2025
GOAT-SLM	Model	-	-	-	-	✓	-	-	-	✓	Speech tokens	Paralinguistic and speaker-aware spoken LM	2025
OpenS2S	Model	-	-	-	-	✓	-	-	-	✓	Speech tokens	Open end-to-end empathetic speech language model	2025
Step-Audio 2	System	✓	-	-	-	✓	-	-	-	✓	Speech-text tokens	Agentic speech interaction with tool use	2025
InteractiveOmni	Model	△	-	-	-	✓	✓	-	-	✓	Audio-visual tokens	Unified audio-visual multi-turn dialogue model	2025
Step-Audio	System	✓	-	-	-	✓	-	-	-	✓	Speech-text tokens	Unified 130B speech-text model with tool calling	2025
Baichuan-Audio	System	✓	-	-	-	✓	-	-	-	✓	Multi-codebook	Text-guided aligned real-time speech generation	2025
Kimi-Audio	Model	✓	-	-	-	✓	-	-	-	✓	Continuous + tokens	Unified audio understanding and generation LLM	2025
Qwen3-Omni	Model	✓	-	-	-	✓	✓	-	-	✓	Continuous + codec	Unified omni-modal Thinker-Talker model	2025
PACHAT	System	△	-	-	✓	✓	-	-	✓	-	Speech frontend + TTS	Persona-aware multiparty speech assistant	2025
VITA-Audio	Model	✓	-	-	-	✓	-	-	-	✓	Interleaved tokens	Fast interleaved cross-modal token generation	2025
AV-Dialog	Model	-	-	-	-	✓	✓	-	-	✓	Audio-visual features	Spoken dialogue model with audio-visual input	2025
Qwen2.5-Omni	Model	✓	-	-	-	✓	✓	-	-	✓	Continuous + codec	Thinker-Talker omni-modal architecture	2025
LLaMA-Omni 2	System	✓	-	-	-	✓	-	-	-	✓	Discrete units	Autoregressive streaming speech synthesis	2025
VoxMind	System	△	-	-	-	✓	-	-	-	✓	Speech tokens	End-to-end agentic spoken dialogue system	2026
ZipVoice-Dialog	Model	△	-	-	-	✓	-	-	-	✓	Continuous latent	Non-autoregressive dialogue generation with flow matching	2026
Interaction Mode: Half-duplex / Controlled Barge-in													
Incremental TT Model	Component	✓	-	-	-	-	-	-	-	-	Incremental features	Incremental task-oriented turn-taking model	2019
Listener-aware BC	Component	△	-	-	✓	-	-	-	-	-	ASR text	Listener-aware backchannel prediction	2020
BPM_MT	Component	△	-	-	✓	-	-	-	-	-	Speech + text	Multi-task backchannel prediction	2021
Device Directedness	Component	△	-	-	-	-	-	-	-	-	Acoustic + context	Classifies whether speech is directed at the device	2022
Acoustic Barge-in	Component	△	✓	-	-	-	-	-	-	-	Acoustic + context	Contextual user barge-in classifier	2022
Freeze-Omni	System	✓	✓	△	△	✓	-	-	-	✓	Continuous + units	Frozen LLM with chunk-level state prediction	2024
Mini-Omni2	System	✓	✓	△	△	✓	✓	-	-	✓	SNAC codec	Multimodal extension with duplex capability	2024
VITA	System	✓	✓	△	△	✓	✓	-	✓	-	Encoder + TTS	Omni MLLM with duplex event/state control	2024
SHANKS	Add-on	✓	△	△	-	✓	-	-	✓	-	Streaming features	Simultaneous hearing and thinking side path	2025
Interaction Mode: Full-duplex													
Duplex Conversation (Outbound)	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / TTS	Modular outbound-agent duplex controller	2021
Duplex Conversation	System	✓	✓	✓	✓	✓	-	-	✓	-	ASR / TTS	Two-channel interaction modeling and control	2022
Synchronous LLM	Model	✓	✓	✓	✓	✓	-	-	-	✓	Speech units	Clock-synchronous Llama 3 dialogue modeling	2024
Moshi	Model	✓	✓	✓	✓	✓	-	-	-	✓	Mimi codec	Helium with temporal and depth Transformers	2024
SALMONN-omni	Model	✓	✓	✓	✓	✓	-	-	-	✓	Continuous embeddings	Codec-free asynchronous text-speech generation	2024
Full-duplex LLM Scheme	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM / TTS	Streaming modular LLM dialogue pipeline	2024
MinMo	Model	✓	✓	✓	✓	✓	✓	-	-	✓	Speech tokens	Multimodal LLM for seamless voice interaction	2025
SALM-Duplex	Model	✓	✓	✓	△	✓	-	-	-	✓	Encoder + codec	Continuous input and codec output with channel fusion	2025
OmniFlatten	Model	✓	✓	✓	✓	✓	-	-	-	✓	Speech-text tokens	Flattened multi-stream GPT with progressive training	2025
Japanese FD System	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM / TTS	Modular Japanese full-duplex dialogue system	2025
FireRedChat	System	✓	✓	✓	△	✓	-	✓	✓	-	ASR / TTS or codec	pVAD controller, interaction module, dialogue manager	2025
FD Dialogue + DST	System	✓	✓	✓	△	✓	-	✓	✓	-	ASR / DST / TTS	Dialogue-state tracking in a Japanese FD system	2025
Chronological Thinking	Model	✓	✓	✓	✓	✓	-	-	-	✓	Synchronized tokens	Chronological reasoning for duplex speech streams	2025
LLM-Enhanced FD-DM	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM-DM / TTS	LLM-enhanced modular dialogue management	2025
LSLM	Model	✓	✓	✓	△	✓	-	-	-	✓	SSL + speech tokens	Listening and speaking channels with middle fusion	2025
Chroma 1.0	System	✓	✓	✓	△	✓	-	-	-	✓	Speech tokens	Real-time end-to-end model with voice cloning	2026
PersonaPlex	Model	✓	✓	✓	△	✓	-	-	-	✓	Neural codec	Duplex model with role prompts and voice cloning	2026

Method	Scope	Interaction Capabilities				Modalities		System Architecture			Speech Interface	Core Design	Release
		Streaming	Barge-in	Concurrent	Backchannel / Overlap	Speech Output	Vision	Cascaded	Semi-casc.	End-to-end			
MiniCPM-o 4.5	Model	✓	✓	✓	△	✓	✓	-	-	✓	Omni-modal stream	Omni-Flow on a shared temporal axis	2026
DuplexSLA	Model	✓	✓	✓	✓	✓	-	-	-	✓	Continuous + discrete	Joint speech, language, and action channels	2026
BayLing-Duplex	Model	✓	✓	✓	✓	✓	-	-	-	✓	Audio tokens	Native duplex modeling in one autoregressive LLM	2026
F-Actor	Add-on	✓	✓	✓	✓	✓	-	-	✓	-	Control states	Behavior-control module for full-duplex models	2026
Hierarchical A-S Modeling	Model	✓	✓	✓	✓	✓	-	-	-	✓	Semantic + acoustic	Hierarchical modality-separated duplex modeling	2026
JoyAI-Talker	Model	✓	✓	✓	△	✓	-	-	-	✓	Semantic + acoustic	End-to-end empathetic full-duplex voice model	2026
SoulX-Duplug	Add-on	✓	✓	✓	△	✓	-	-	✓	-	Streaming ASR	Plug-in semantic state prediction module	2026

Abbreviations: ASR, automatic speech recognition; TTS, text-to-speech; SSL, self-supervised learning; LLM, large language model; VAD, voice activity detection.